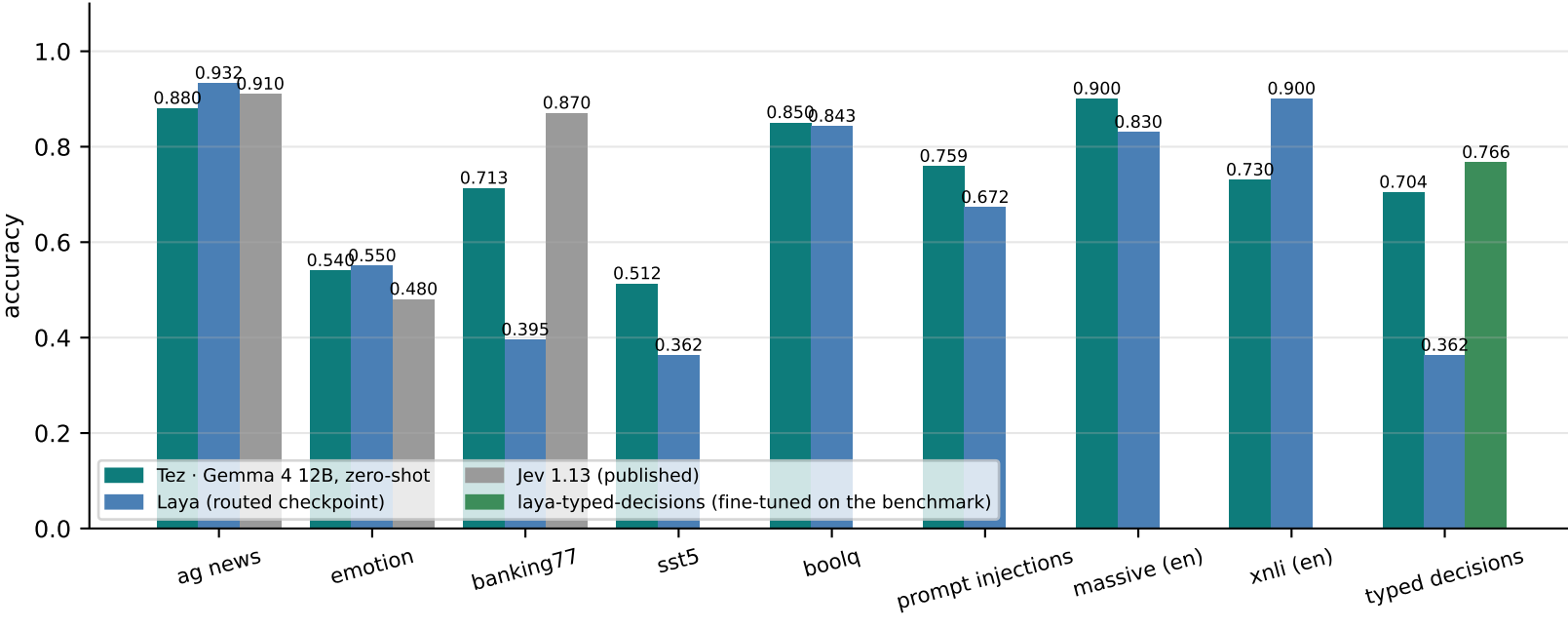
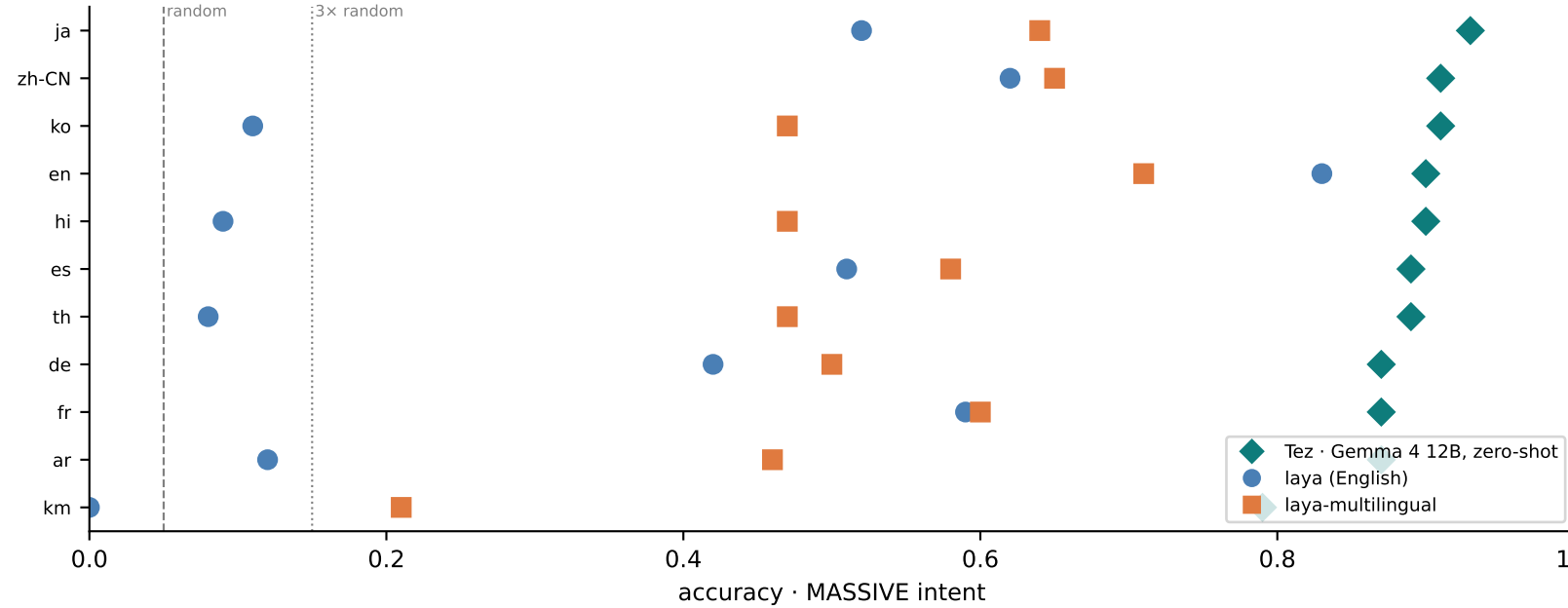


Tez — every measurement, one page. Zero-shot frozen Gemma 4 12B Q8_0 — one forward pass — RTX 5080 laptop (16 GB)
Laya checkpoints ran on the SAME rows and GPU; Jev figures are third-party published, never measured here.

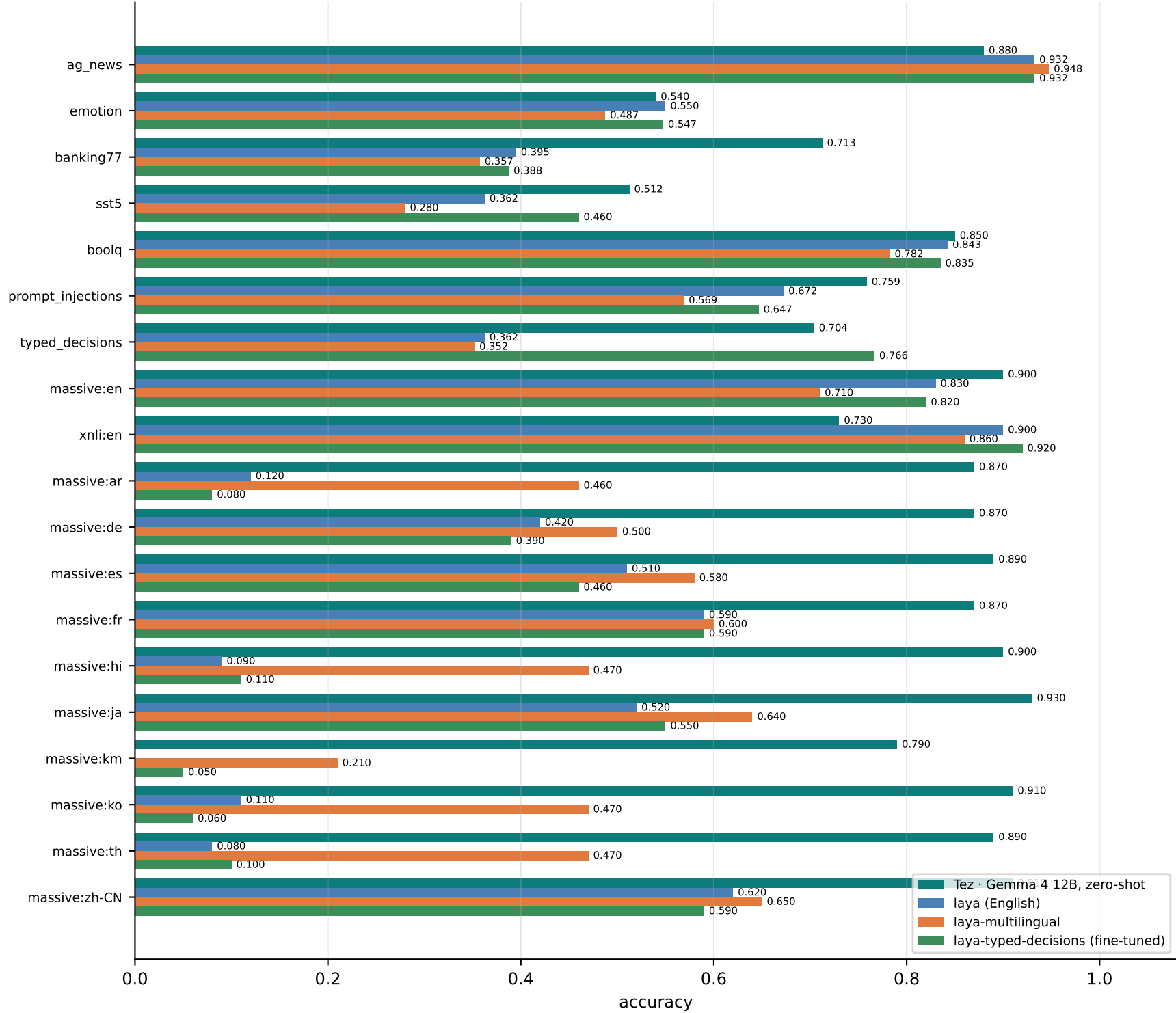
A · Accuracy on Laya's public datasets — same rows, same machine



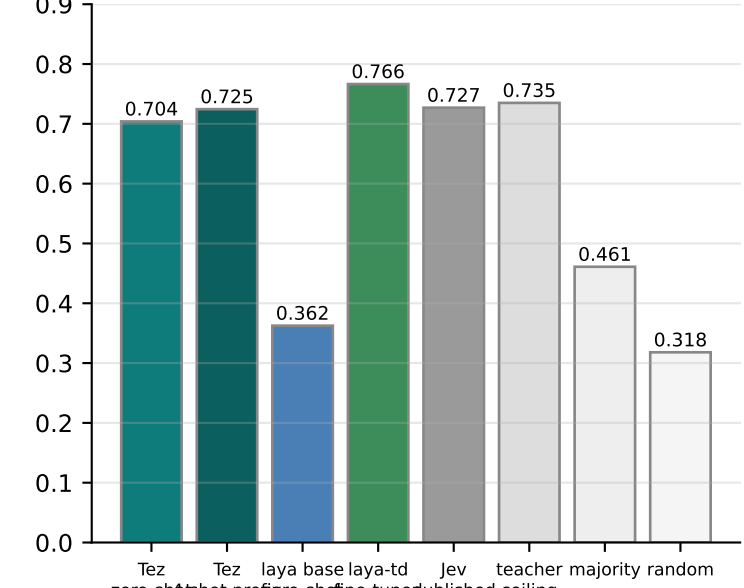
C · One frozen decoder vs two trained encoders, per language



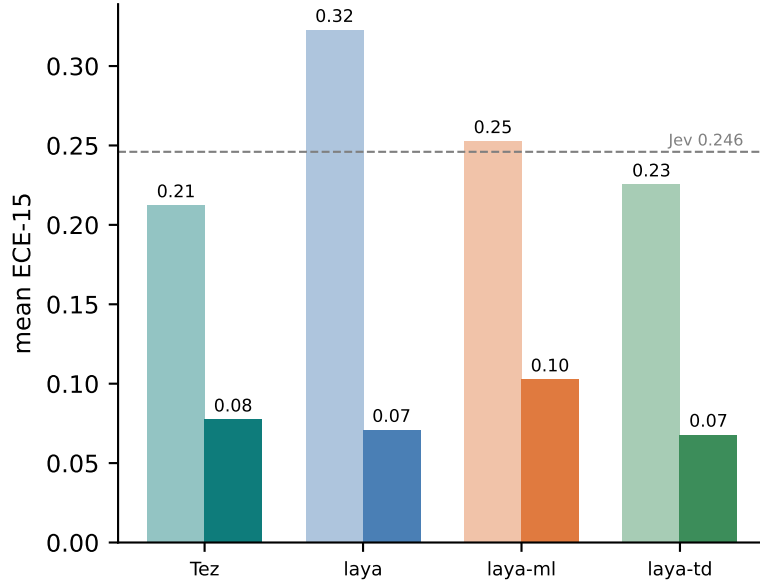
B · Every task, every checkpoint (MASSIVE: 20 options, 100/lang)



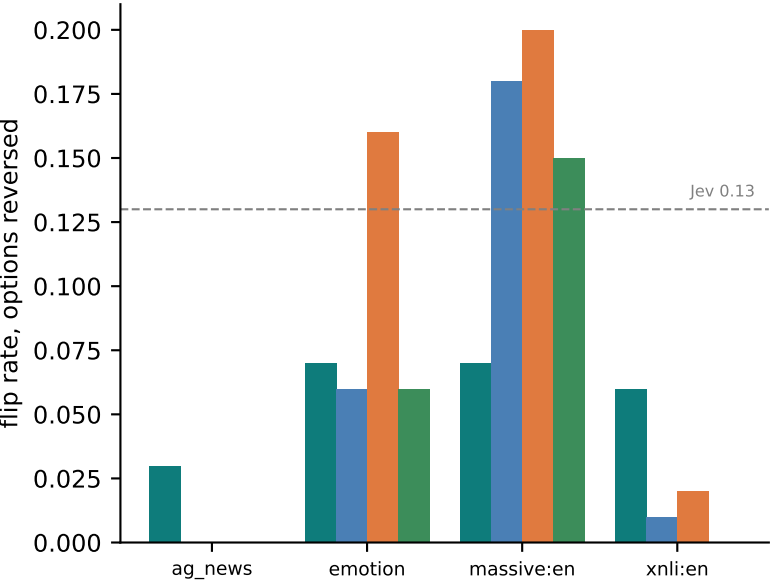
D · typed-decisions (2,000 decisions)



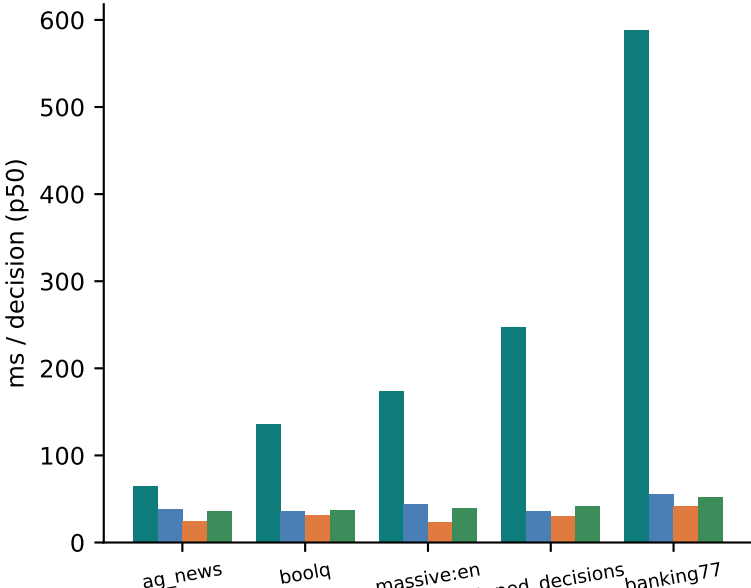
E · Calibration: shipped (light) → per-task refit (dark)



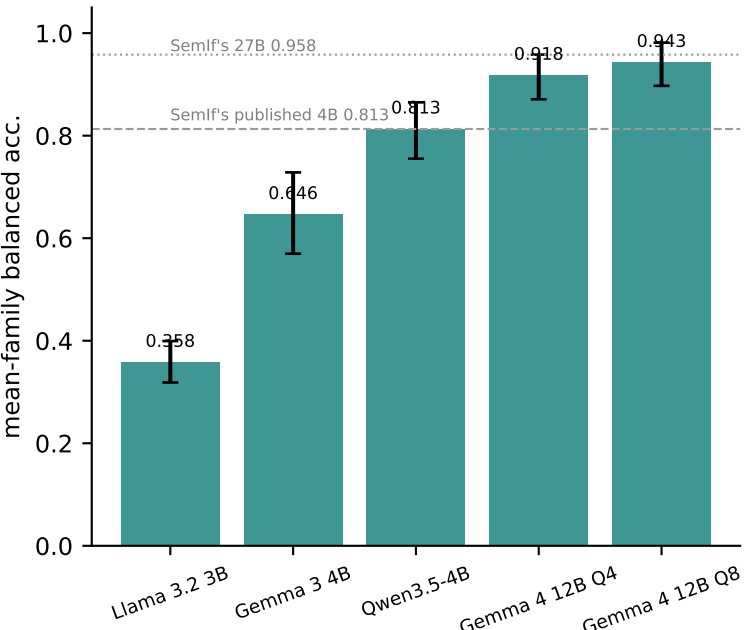
F · Option-order robustness (first 100 rows)



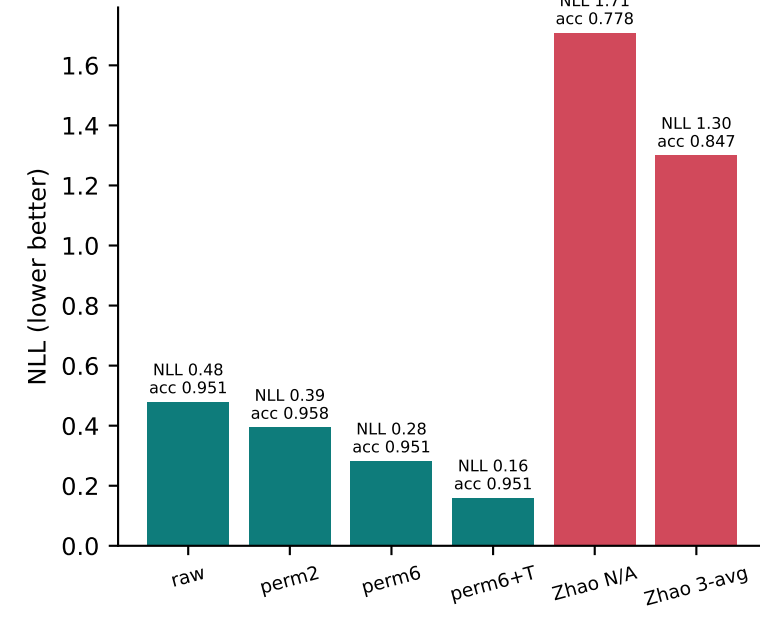
G · Latency, same GPU (Tez over HTTP, state uncached)



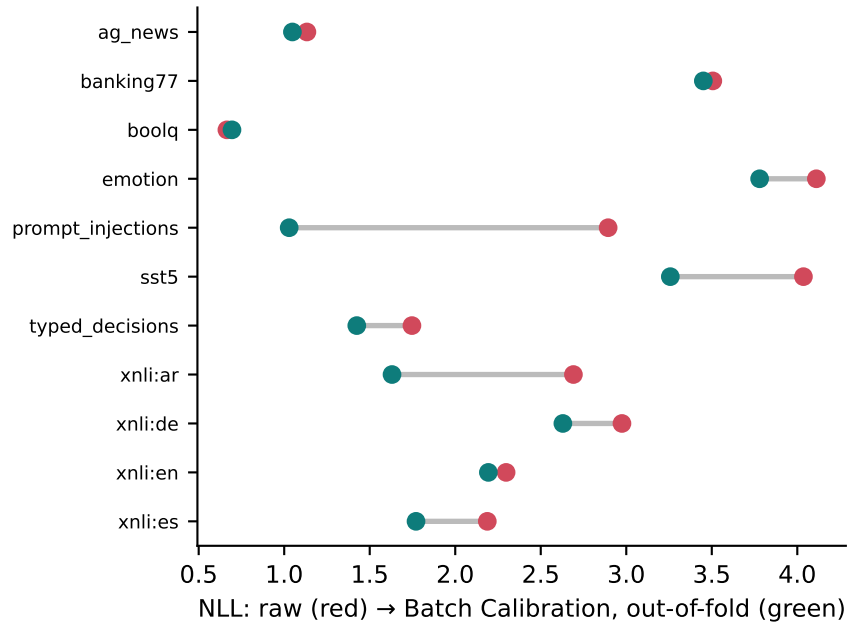
H · Scale-gated symbol binding (SemIf, 95% CI)



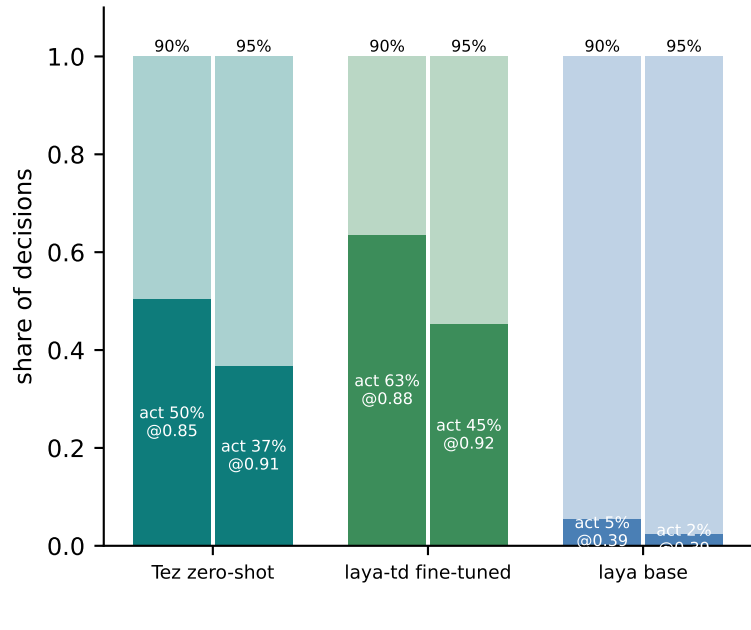
I · Calibration recipes, Gemma 4 12B Q8



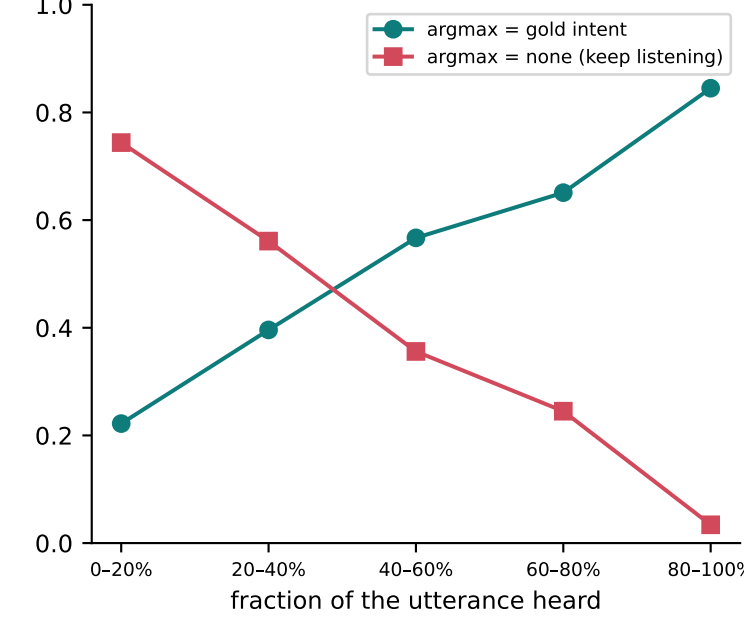
J · Batch Calibration (Zhou 2023), NLL



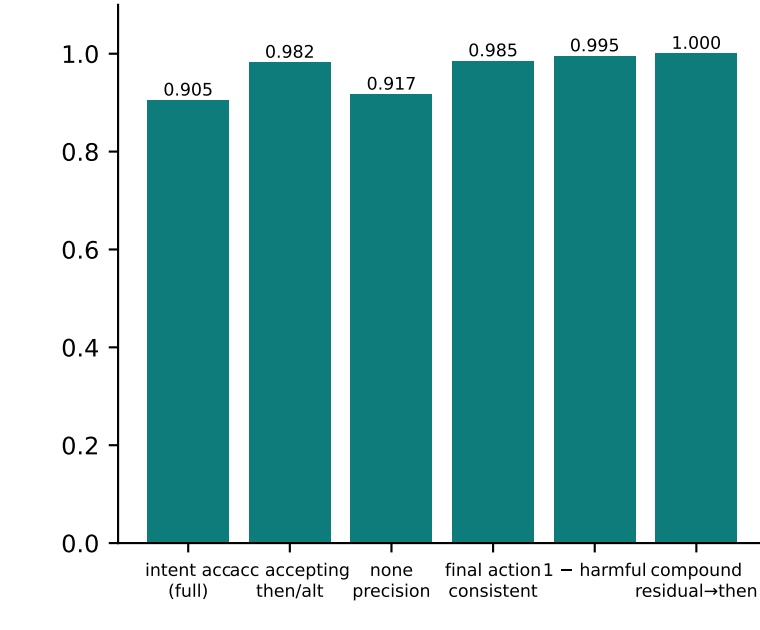
K · Conformal act / escalate (typed-decisions)



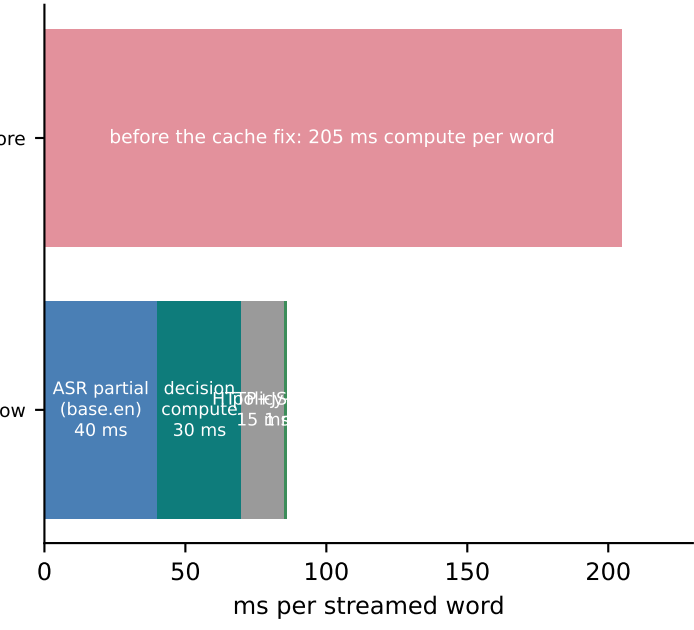
L · Voice: partial-transcript predictions



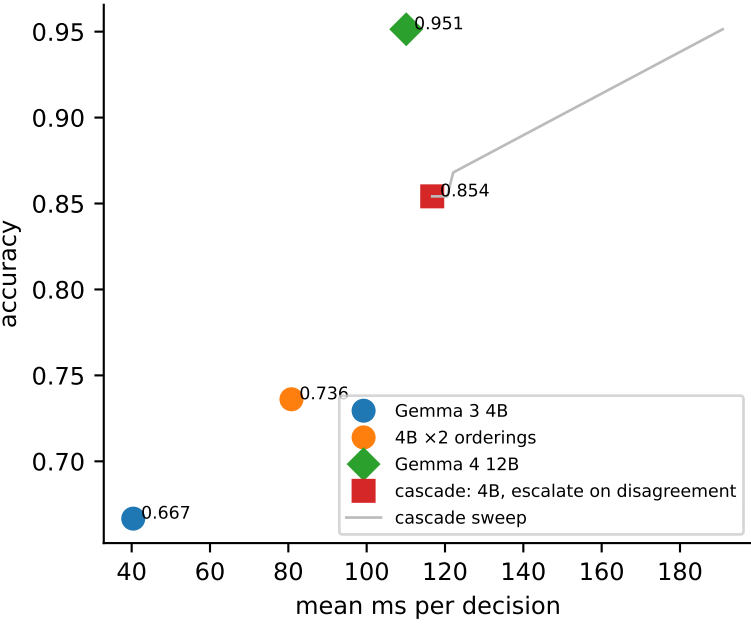
M · Voice → action outcomes (220 commands)



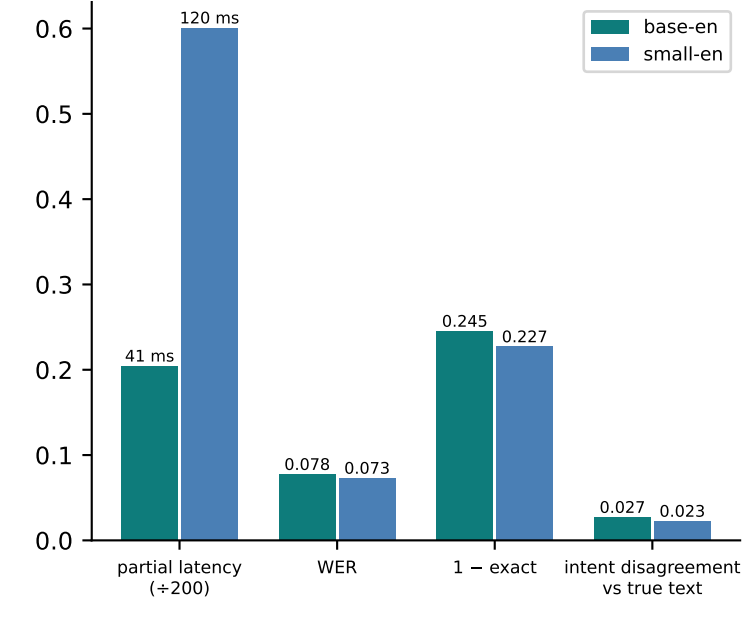
N · Per-word budget (~85 ms word → action)



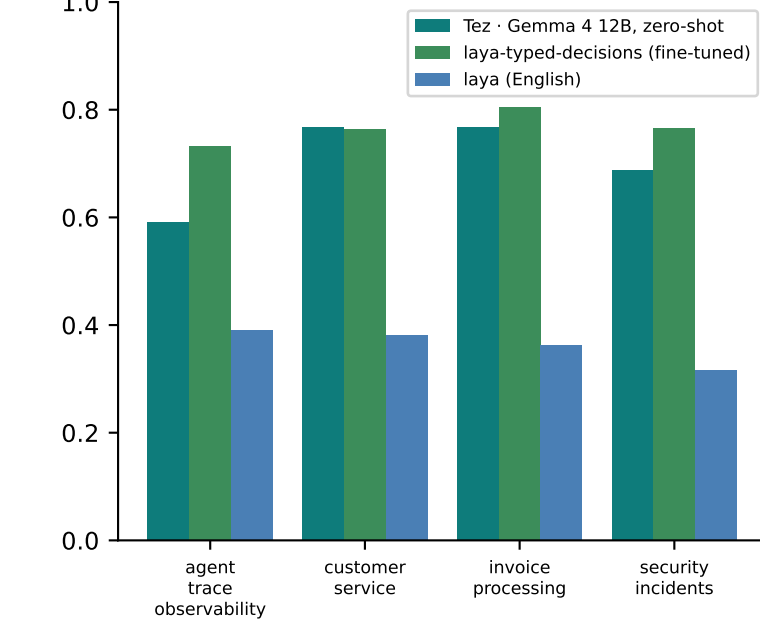
O · Small → large cascade: not worth it



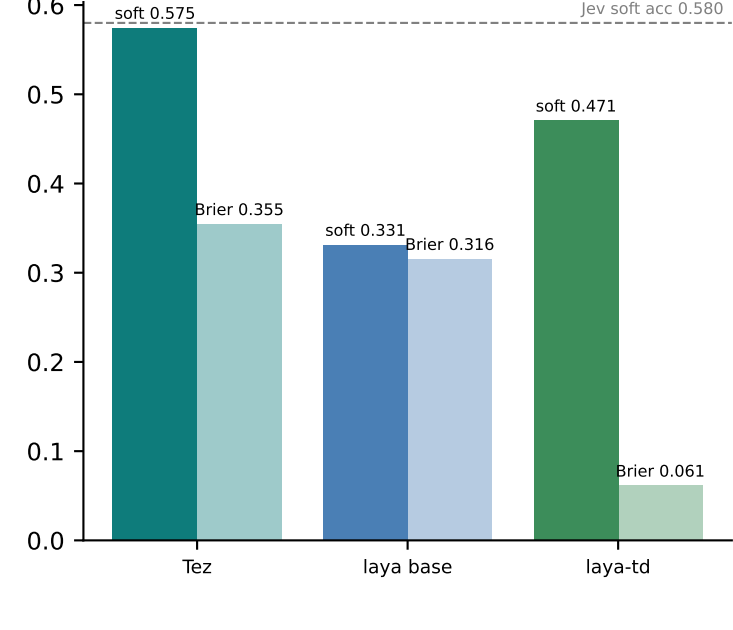
P · ASR stage (faster-whisper, synthetic speech)



Q · typed-decisions by workflow



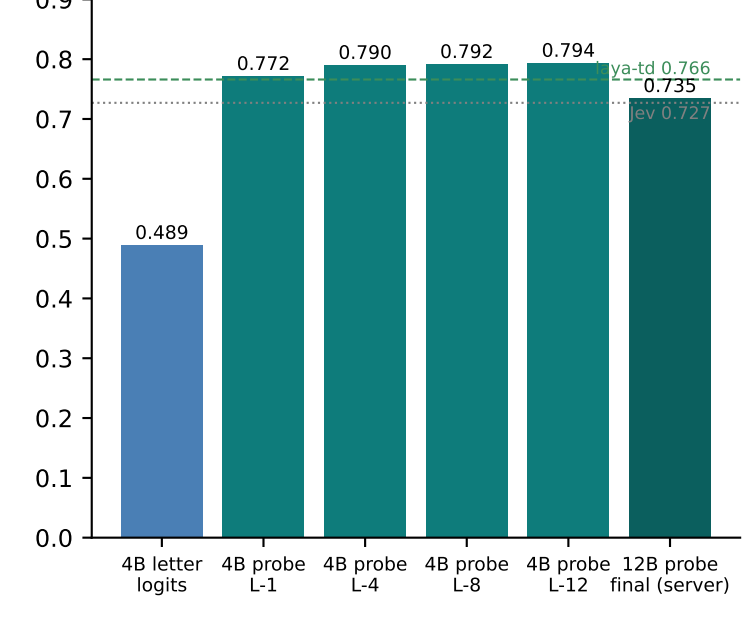
R · Teacher-distribution match (soft acc ↑, Brier ↓)



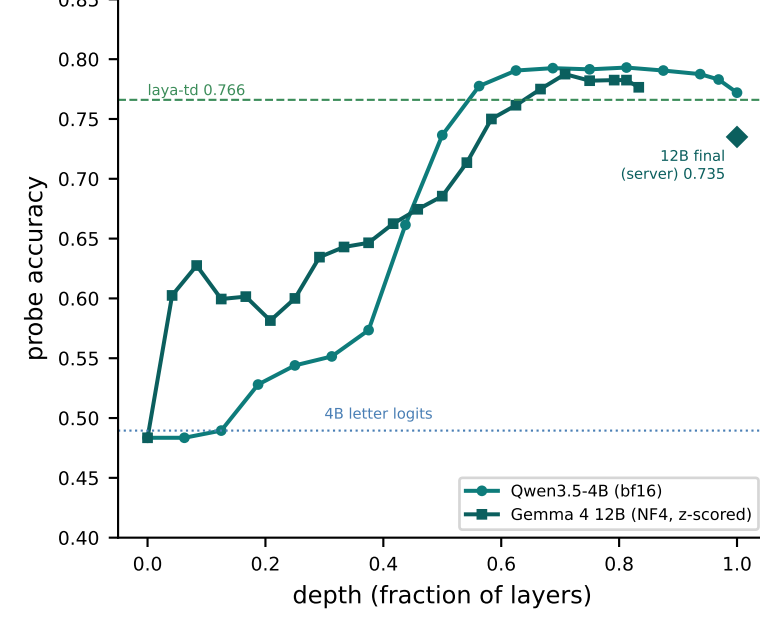
Headline
typed-decisions zero-shot 0.704 (Laya base 0.36 · Jev 0.727 · Laya fine-tuned 0.766)
MASSIVE, 11 languages 0.885 macro (laya-multilingual 0.524 · Khmer 0.790 vs 0.000)
Banking77, 77 options 0.713 (Laya 0.395)
SemIf authored144 0.943 (SemIf 4B 0.813)
voice: per streamed word 30 ms compute - 1 harmful action / 198 order flip @ 20 options 0.07 (Laya 0.15-0.20 · Jev 0.13)
conformal 90% coverage act on 52% @ 0.853 accuracy
probe on frozen 4B (L26) 0.794 · 50 rows/question 0.741

Where others win: Laya on tasks in its training mix (AG News, NLI) and on single-question latency; Jev on Banking77 (0.070).

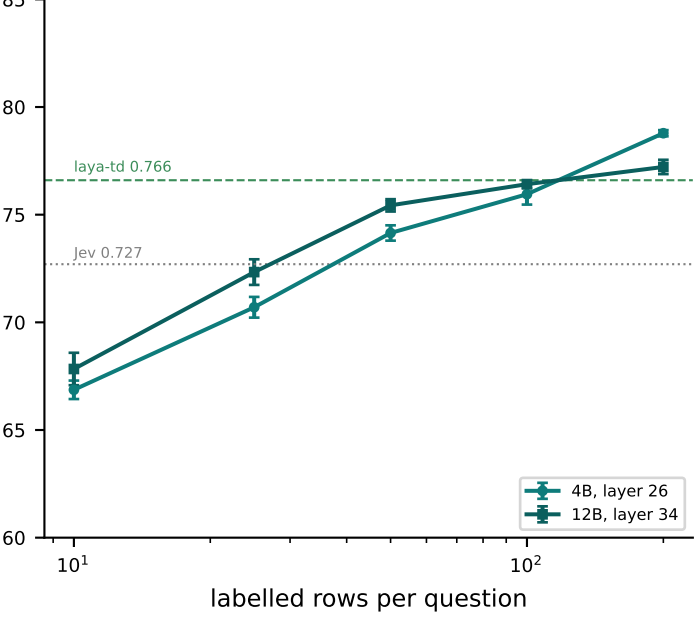
S · Probes on frozen hidden states (typed-decisions)



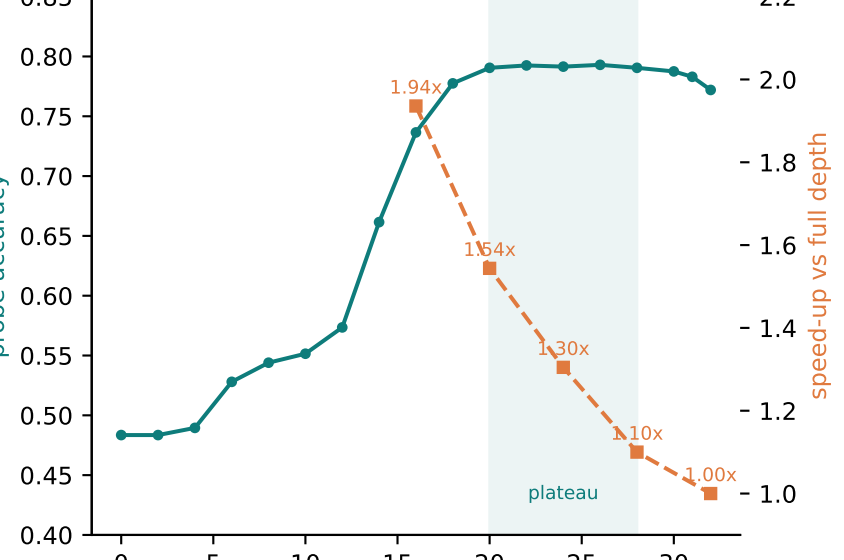
T · Probe accuracy by depth, 4B vs 12B



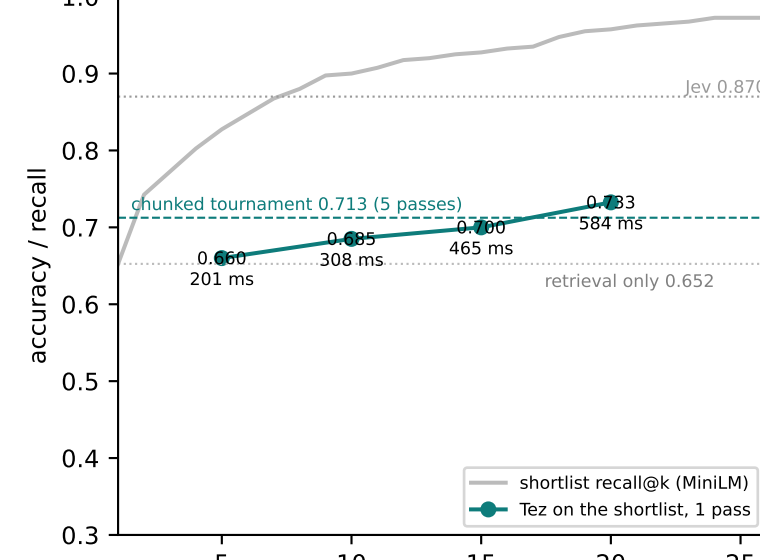
U · Probe data efficiency



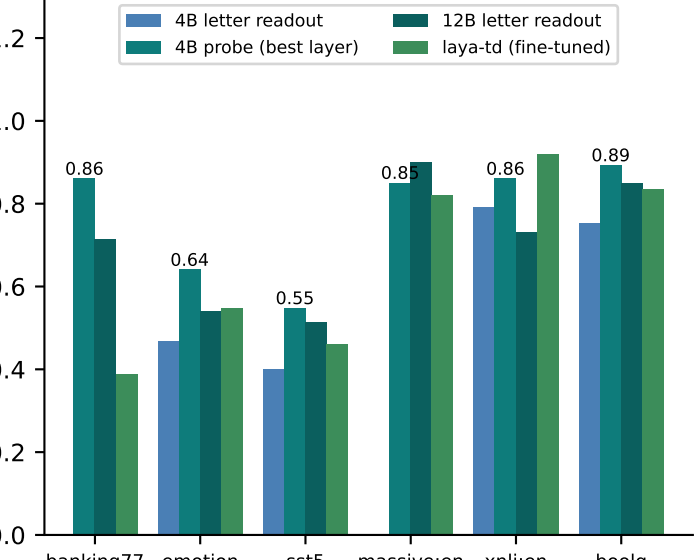
V · Early readout: probe accuracy vs prefill speed-up (4B)



W · Banking77: shortlist, then decide



X · One probe per task, 2,000 rows



Y · Option elimination: no gain

